Robot / rover

DO | RASPBERRY PI | OUTPUT | MOVEMENT

What does it do?

A two-wheel rover uses a separate geared motor for each wheel. A caster or smooth skid supports the other end.

Drive both wheels forwards to move; change their speeds or directions to turn. A motor driver handles their power.

The Robot class joins the two motors into one object. This example uses a breadboard-mounted L293D chip and a separate motor battery pack.

Uses



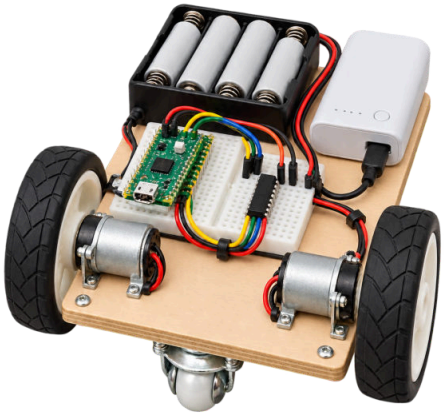
Robot vacuum cleaner

DETECT Sense a nearby obstacle → **DECIDE** Is the path clear? → **DO** Drive on or turn away



Sock delivery turtle

DETECT Read delivery button → **DECIDE** Is a delivery requested? → **DO** Drive the sock parcel



Drive wheels
One geared motor per side

Motor driver
L293D, a 16-pin breadboard chip

Caster or skid
Supports the chassis

Power
Separate motor and USB supplies

Raspberry Pi Foundation

Code

Raspberry Pi

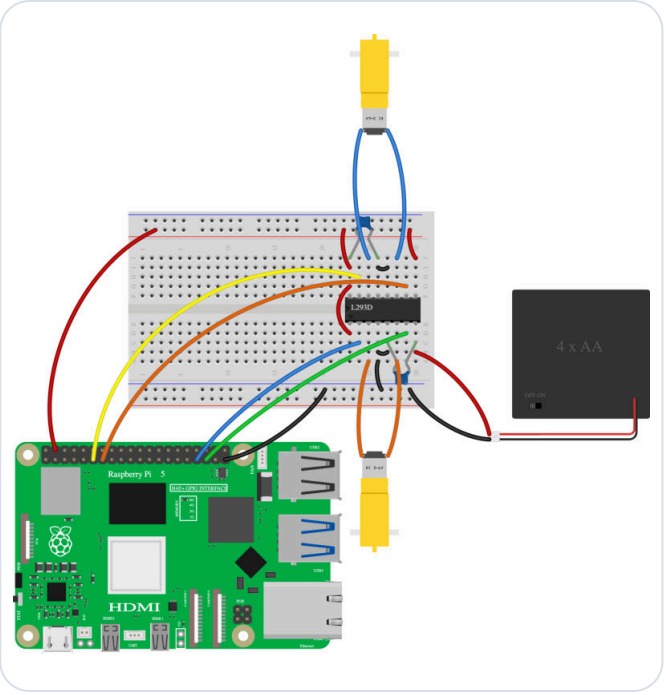
```
from gpiozero import Motor, Robot
from time import sleep

robot = Robot(left=Motor(13, 19),
              right=Motor(17, 27))
try:
    robot.forward()
    sleep(1)
finally:
    robot.stop()
```

Test with wheels lifted. Reverse a motor by swapping its leads with power off. Try left(), right(), backward() and stop().

Wiring

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GPIO13/19 (Pi pins 33/35) → L293D pins 2/7; **GPIO17/27** (Pi pins 11/13) → 15/10. **5 V** (Pi pin 4) → 1, 9, 16. **GND** (Pi pin 39) → 4, 5, 12, 13.